Supporting evidence

S1. Eligibility and fixed model specification

Stage / exclusion reason	Count
ok	6759
previous_rating_or_curated_eeg_unusable	2252
current_rating_or_curated_eeg_unusable	541
first_video_in_session	174
Final paired identifiers	53
Final J1 / J2 transitions	2294 / 2260

The exclusion categories are applied sequentially: a row can have multiple defects but is assigned its first eligibility reason. The predecessor is chosen before exclusions; an unavailable intermediate video is never skipped. Recorded days replace inconsistent raw label days. The whole-EEG quality rule requires usable frontal as well as posterior electrodes.

Frozen setting	Value
seed	20261109
primary_cohort	2
minimum_pairs_per_day	20
precision_replicates	100
variance_bootstrap_draws	999
predictive_bootstrap_draws	10000
prediction_folds	5
population_alpha	10
personal_penalty	20
source_job_suffix	20261108T230726-28480a
reference_gap_s	15.957

Random-effect covariance models individual standardized lagged-arousal contrasts, not raw emotions or neural states. Full off-diagonal sampling covariance is retained. HC3 denominators have a numerical floor of 0.05; substantial covariance indefiniteness is rejected, and numerical adjustments are recorded in the result JSON. The latent-correlation profile is evaluated on a finite grid and a boundary estimate is not interpreted as a precise correlation.

Scientific source: 5b079a836da54652240c00c43555a1624400ae0e
Frozen at UTC: 2026-11-09T12:37:36.537708+00:00
Scientific code SHA256: 6031e4f9ecfa7d491352b3f7651f6f20ad8655fd8ab9ecb9eb1ac7cf12b8a5cb

S2. Prespecified sensitivity analyses

Specification	IDs / pairs	Covariance	Correlation profile
sampling half	53 / 4554	0.0132	[0.0000, 0.9950]
sampling double	53 / 4554	4.73e-10	[-0.9950, 0.9950]
without count moderator	53 / 4554	0.0028	[-0.9950, 0.9951]
hc1 no serial kernel	53 / 4554	0.0094	[-0.9950, 0.9951]
minimum 30	46 / 4105	0.0040	[-0.9950, 0.9951]
common j1 scales	53 / 4554	0.0057	[-0.9950, 0.9951]
stable j1 scales	53 / 4554	-	identical primary grid and scales
j1 gap iqr	17 / 969	-9.99e-08	[-0.9950, 0.9950]

These are sensitivity analyses, not independently selected discoveries. Common normalization changes the operational contrast's units; covariance magnitudes across that specification are not directly interchangeable. The gap-IQR analysis reapplies the twenty-pair-per-day support rule after restriction. An unidentified or insufficient design is reported without forcing an estimate.

S3. Model-based uncertainty and predictive uncertainty

The covariance interval and null probability use parametric draws from the fitted normal-normal models, holding estimated sampling covariance fixed. They account for correlated slope errors under that specification; they are not distribution-free and do not correct measurement error in lagged predictors. The prediction intervals instead resample the 53 identifiers' paired errors and keep the frozen grouped forecasts fixed. Both condition on the released video catalogue.

Saving and reading the forecasts

Cross-validation reports in skore 0.29.1 reconstruct source-bound splits while loading. Re-reading therefore restores the exact private feature table, recorded events and J2 query CSV, plus report datasets, at their original /out paths. A unit test forbids forecaster fitting during report retrieval. The completed-run audit checks cached predictions, registered fold metrics, global equal-person errors, paired intervals and the saved REML objective without re-optimizing any model. Generic skore checks did not complete on the constant-day query metadata and reported an array-dimensionality error. This is recorded as unavailable, not as successful automated leakage or overfitting validation; cached metrics and targeted invariants were checked separately.

S4. Pre-effect synthetic precision diagnostics

All numbers on this page come from simulated slopes under the actual predictor geometry. They are not measured subjective or EEG associations. Assumptions were difference-noise variance 2 and event-order serial correlation 0.3; 100 replicates were generated per scenario. Correlation has no generating meaning when random-effect variance is zero.

Scenario / SD	True rho	Estimated covariance: 2.5%, median, 97.5%
Random SD 0.00	undefined	-0.0105, 1.31e-07, 0.0127
Random SD 0.15	0.0	-0.0249, -3.55e-04, 0.0191
Random SD 0.15	0.6	-0.0136, 0.0103, 0.0297
Random SD 0.30	0.0	-0.0472, 0.0011, 0.0364
Random SD 0.30	0.6	0.0090, 0.0493, 0.1084
Random SD 0.50	0.0	-0.0812, 0.0028, 0.0909
Random SD 0.50	0.6	0.0516, 0.1406, 0.2330
Shared scale only	undefined	-0.0083, -1.43e-08, 0.0093

With small true variation, point correlations frequently approach the optimization boundary even though covariance is modest or null. The scale-only scenario shows why a raw test-retest scatter can be misleading. Simulated precision supports an explicit assessment of identifiability and uncertainty; it is not a guarantee of power for arbitrary biological effects.

Evidence files

The accompanying results folder contains empirical aggregates, the exact freeze, structural/simulation evidence and the executed read-only audit. Source code, input fingerprints and software versions allow authorized HABS users to reproduce the analysis. Raw individual annotations, source features and saved report datasets are not distributed with this PDF.